

Shark Evolution Experiment

GA + RL run — 2026-06-09T10:56:57

Summary

Summary	
Generations	1000
Population (sharks/gen)	30
Best fitness	115.600
Final mean fitness	3.221
Analytical fitness (best genome)	-0.0240
Runtime	118.5 s

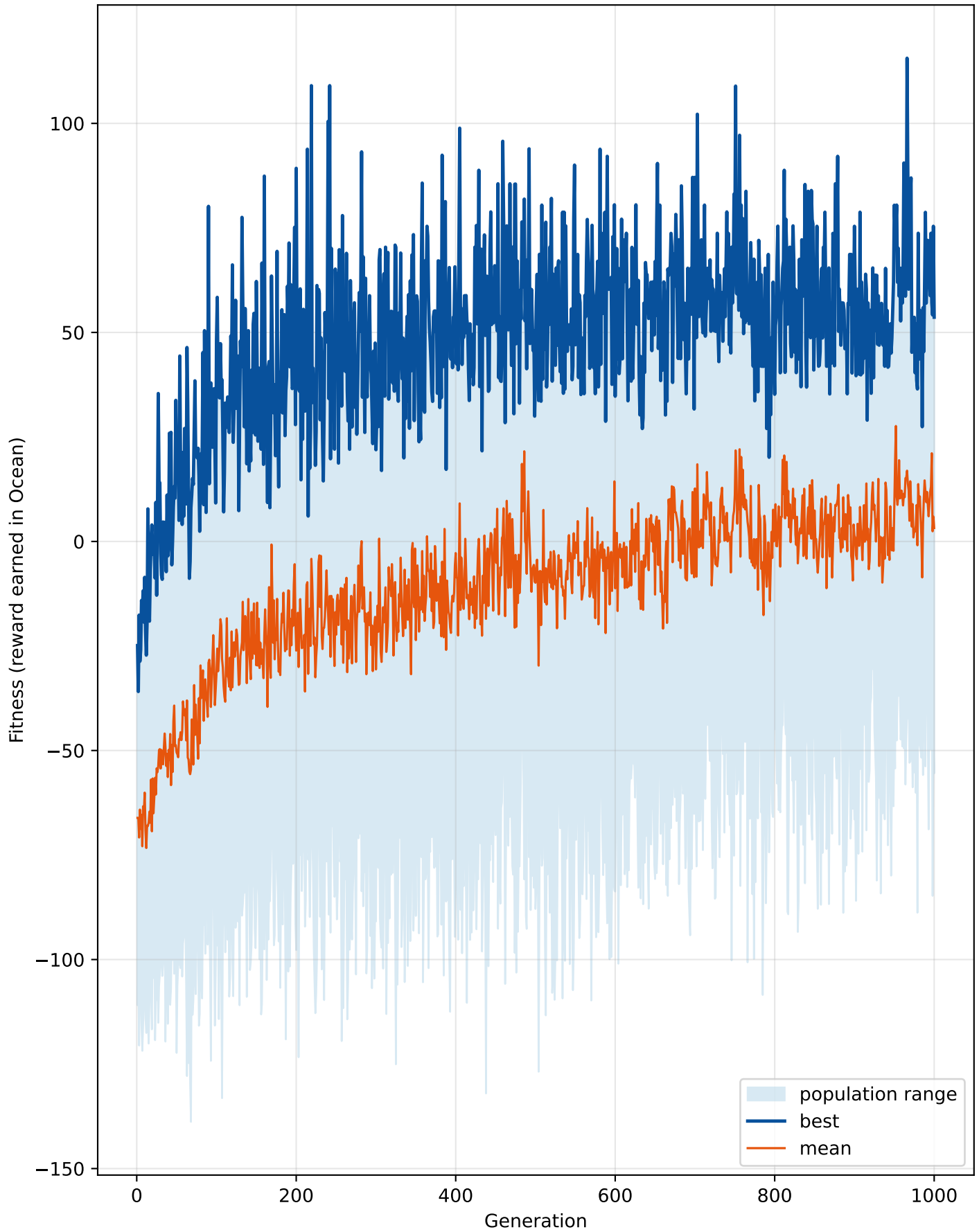
Run parameters

Run parameter	Value
Max steps per life	200
Lives averaged per genome	3
Mutation rate	0.2
Random seed	0
Run timestamp	2026-06-09T10:56:57

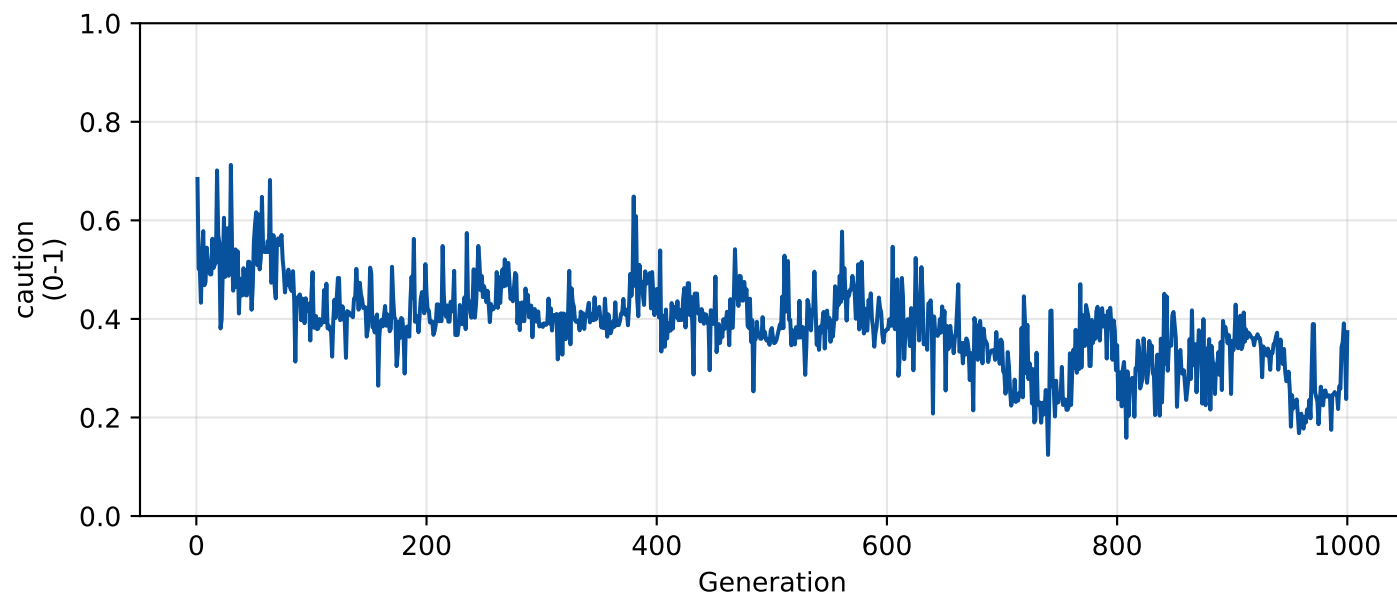
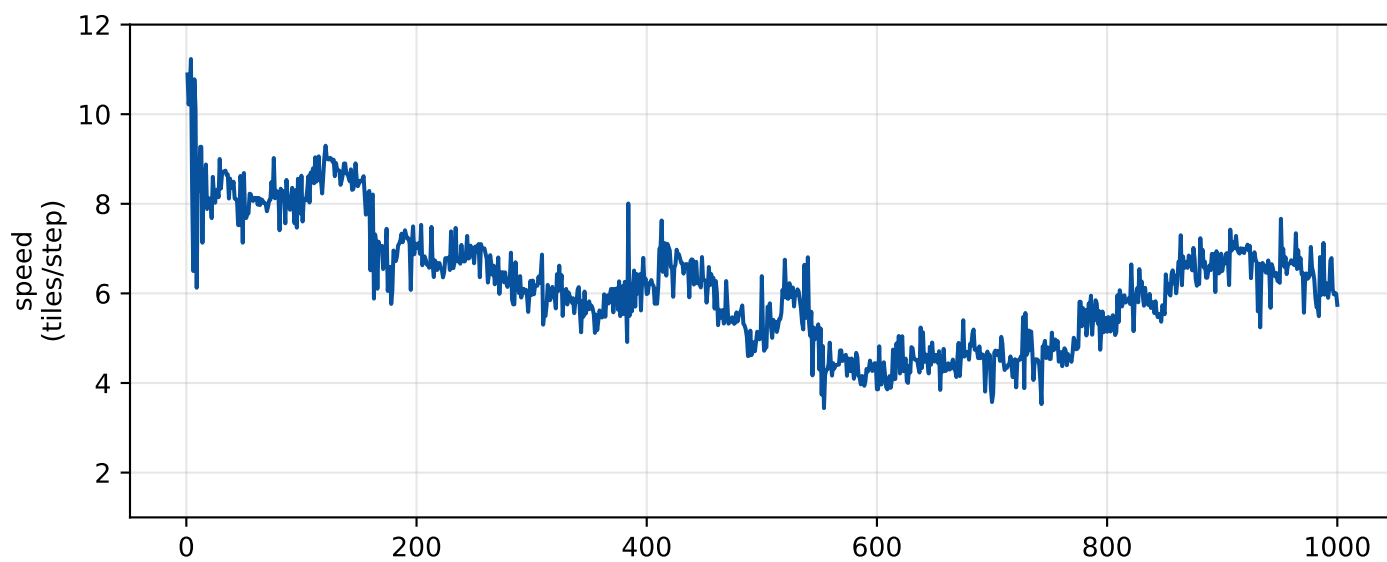
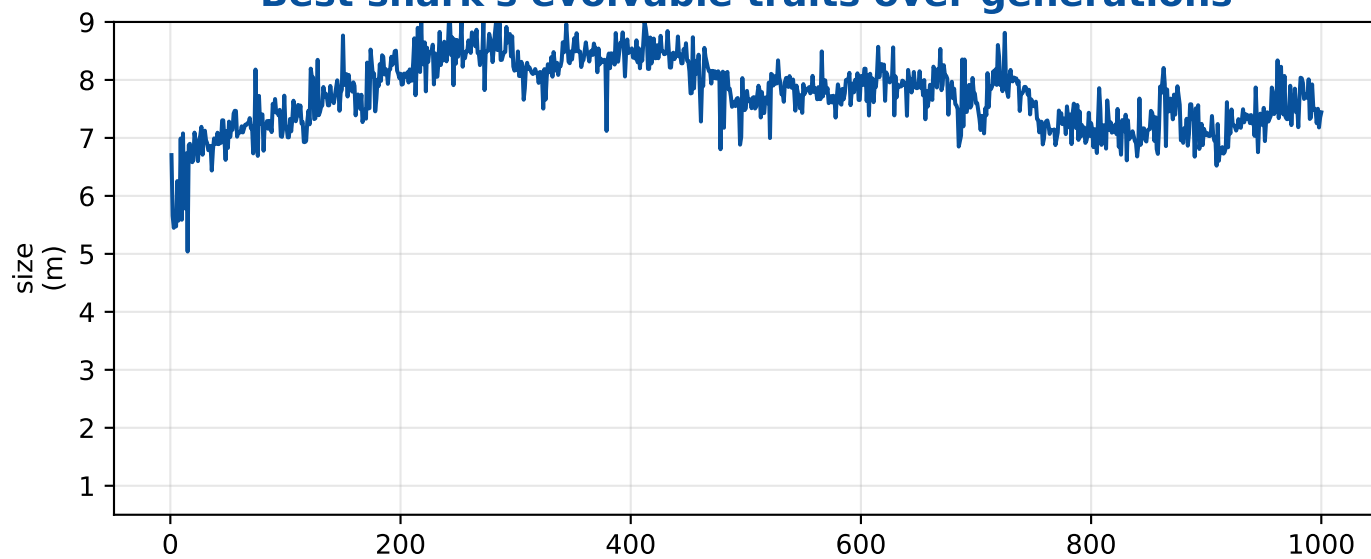
Shared brain (Q-learning)

Shared brain (Q-learning)	Value
States learned	237
Final epsilon (explore rate)	0.050
Epsilon min / decay	0.050 / 0.9970
Alpha (learning rate)	0.100
Gamma (discount)	0.950

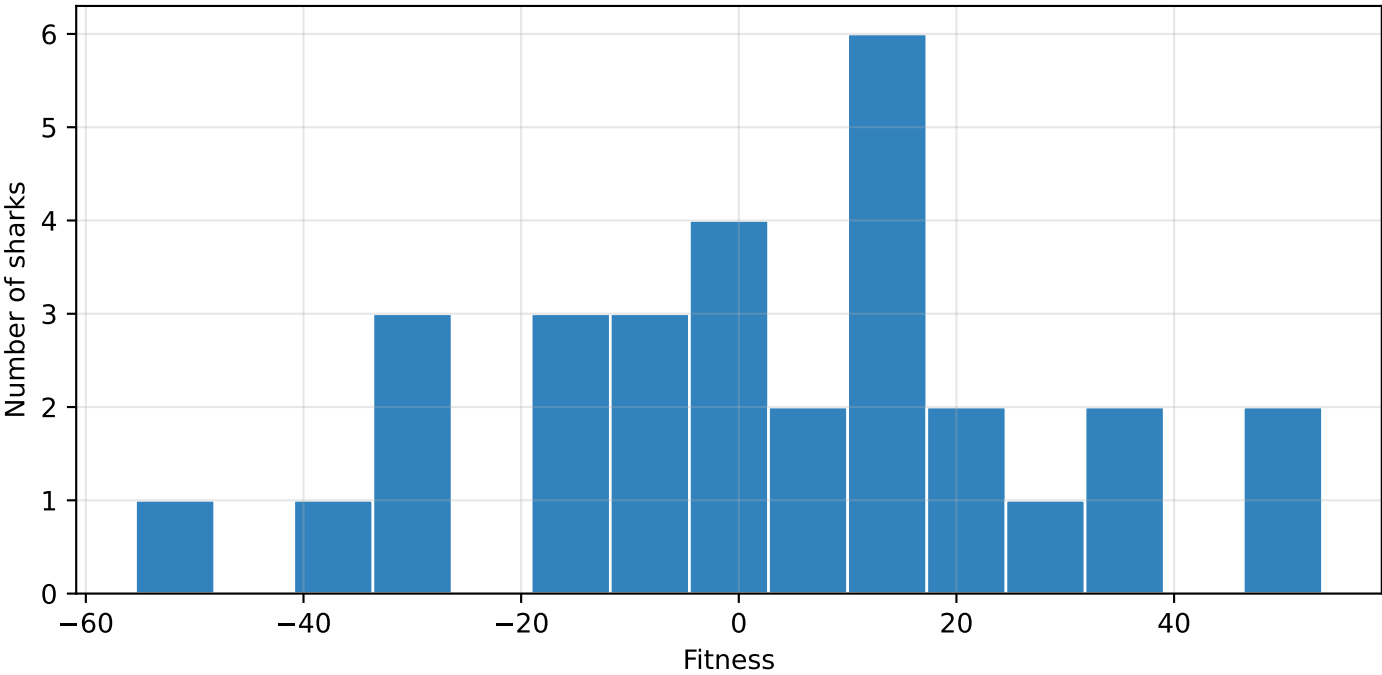
Fitness over generations



Best shark's evolvable traits over generations



Final-generation fitness distribution (gen 1000)



Best genome found (judged by reward earned in the Ocean)

Trait	Value	Unit	Evolved?
size	7.502	m	yes
speed	6.964	tiles/step	yes
caution	0.202	0-1	yes
gestation_period	12.000	months	no (fixed)
aggression	0.500	0-1	no (fixed)
field_of_perception	5.000	tiles (radius)	no (fixed)
color_hue	205.000	degrees (HSV hue)	no (fixed)

Per-generation summary (showing 31 of 1000 generations, every 34)

Gen	Size	Speed	Caution	Best	Mean	Min	Max
1	6.704	10.887	0.684	-24.900	-66.113	-110.900	-24.900
35	6.862	8.642	0.465	4.600	-45.940	-86.400	4.600
69	7.342	7.990	0.442	11.067	-42.572	-94.800	11.067
103	7.010	8.098	0.382	34.567	-28.244	-114.233	34.567
137	7.561	8.901	0.441	42.467	-19.760	-77.767	42.467
171	8.299	6.543	0.443	30.800	-24.214	-77.167	30.800
205	8.055	6.628	0.395	60.433	-13.788	-75.000	60.433
239	8.455	7.083	0.402	55.033	-14.516	-74.733	55.033
273	7.825	6.455	0.441	57.100	-10.367	-56.933	57.100
307	7.659	6.232	0.391	16.967	-28.816	-97.167	16.967
341	8.649	6.141	0.433	59.067	-9.046	-75.700	59.067
375	8.305	5.635	0.387	50.133	-5.656	-77.767	50.133
409	8.465	5.768	0.360	51.967	-14.756	-77.633	51.967
443	8.444	6.705	0.409	69.933	-1.339	-76.267	69.933
477	8.087	5.348	0.451	67.133	-4.762	-52.267	67.133
511	7.900	5.322	0.528	45.933	-11.108	-53.367	45.933
545	7.800	5.238	0.414	55.633	-9.933	-65.267	55.633
579	7.684	4.355	0.439	36.967	-19.821	-93.467	36.967
613	7.879	4.065	0.483	65.433	-7.414	-76.333	65.433
647	7.883	4.407	0.372	59.233	4.101	-61.233	59.233
681	7.848	4.877	0.396	57.667	0.227	-67.000	57.667
715	8.147	4.403	0.250	62.000	16.600	-33.267	62.000
749	7.624	4.791	0.231	83.100	7.263	-75.167	83.100
783	6.894	5.277	0.425	63.800	-6.522	-64.800	63.800
817	7.269	5.836	0.297	62.000	12.092	-89.167	62.000
851	7.151	6.425	0.361	49.133	0.464	-59.400	49.133
885	7.065	6.706	0.248	57.000	-1.482	-57.533	57.000
919	6.838	6.901	0.346	41.933	-4.742	-54.900	41.933
953	7.442	6.997	0.218	80.467	10.648	-47.600	80.467
987	7.725	6.049	0.246	45.400	10.181	-30.233	45.400
1000	7.441	5.744	0.373	53.633	3.221	-55.433	53.633